

TOPIC : GARBAGE DETECTION IN PUBLIC AREAS USING PYTHON

#code:

```
import cv2
from ultralytics import YOLO
import time

# Load YOLO model
model = YOLO("yolov8n.pt")

# Open CCTV video file
cap = cv2.VideoCapture("public_cctv_stream.mp4")

last_alert_time = 0
ALERT_INTERVAL = 5

while cap.isOpened():

    ret, frame = cap.read()

    if not ret:
        print("Video ended")
        break

    # Run detection
    results = model(frame)

    # Draw detections
    annotated_frame = results[0].plot()

    # Save current frame
    cv2.imwrite("output.jpg", annotated_frame)

    # Alert system
    boxes = results[0].boxes

    if len(boxes) > 0:
```

```
current_time = time.time()
```

```
if current_time - last_alert_time > ALERT_INTERVAL:  
    print("[ALERT] Object detected!")  
    last_alert_time = current_time
```

```
cap.release()
```

```
print("Processing completed")
```

#OUTPUT:

